Do group-affiliated firms time their equity offerings?

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Abstract

Using IPO and SEO data from the Indian capital market, we examine whether group-affiliated firms are able to time their equity issuance relative to stand-alone firms. Group-affiliated firms have access to internal capital markets, which, we argue should allow these firms to raise external capital at an opportune time. Using both direct and indirect measures of market timing, the main finding of our paper is that there are significant difference in how group-affiliated and stand-alone firms raise capital through IPOs and SEOs. We find that stand-alone firms raise more capital and time their IPO offerings whereas group-affiliated firms raise more capital and time their SEO offerings. Consistent with the notion of market timing, we find that the long run underperformance is pronounced for stand-alone IPO and group-affiliated SEO offerings.

JEL Classification: G31, G32

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1 Introduction

The opportunistic timing of equity issuance has received considerable attention in the academic literature. Importantly, empirical evidence that managers attempt to sell highly priced shares is convincing both for initial public (IPOs) and secondary equity (SEOs) offerings. Using the long-run performance of IPOs and SEOs, Ritter (1991) and Loughran and Ritter (1995) attribute the decline in the performance in the post issuance period to the market timing attempts of the issuers. Similarly, Rajan and Zingales (1995), Pagano et al. (1998) and Baker and Wurgler (2002), among others, demonstrate evidence of market timing by documenting a positive relation between firms' market valuations (market-to-book ratio) and equity issues. Alti's (2006) further shows that hot equity markets have a significant and positive effect not only on the amount but also on the number of equity issues. In this paper, we extend this literature by assessing market-timing abilities of group-affiliated and stand-alone firms in the context of an emerging market. While the extant literature has extensively analyzed market timing, much less in known about the market timing ability of group-affiliated firms relative to stand-alone firms.

A unique feature of most emerging markets is the presence of large business groups. Business group affiliates function as separate legal entities but are connected with the parent group through common equity ownership (see, La Porta et al., 1999). Previous studies highlight the role of business groups in reducing the risk of member firms by sharing (Khanna and Yafeh, 2005) and helping the financially constrained member firms in raising external equity (Hoshi et al., 1991). Although business groups have also been associated with the tunneling phenomenon (Bertrand et al., 2002), a number of studies emphasize the importance of groups for member firms in raising external capital. For instance, Hoshi et al. (1991) find that investment of group-affiliated firms are less sensitive to their liquidities as compared to unaffiliated firms and attribute this to the advantage business group firms have in terms of access to external funding at lower cost of capital. In a similar vein, Almeida et al. (2015) find significant increase in the level of investment in group-affiliated firms after the Asian financial crisis relative to investment in unaffiliated firms.

An important characteristic that differentiates group-affiliated from stand-alone firms is the presence of internal capital market. The internal capital market is a channel through which capital is allocated within various firms or divisions of a parent business group. This feature is important from a market timing perspective as group-affiliated firms can leverage the internal capital markets to overcome funding constraints, which is not possible for stand-alone firms. Further, the availability of internal capital markets should allow group-affiliated firms to raise funds not only at better terms but also at an opportune time. Thus, we argue that because of the presence of internal capital markets, group-affiliated firms should behave differently from stand-alone firms when raising capital from external markets. In line with this argument, Masulis et al. (2016) find that the proportion of group firms going public peaks in the weakest IPO markets and declines in hot markets.

Hence, the main objective of the study is to examine if there are any difference in the market timing of equity issuance between group-affiliated and stand-alone firms. To investigate this issue, we study both IPOs and SEOs in the context of the Indian market, which provides a suitable setting for at least two reasons. First, Indian business landscape is dominated by the presence of several large business groups (Khanna and Palepu, 1996; and Marisetty et al. 2008) that have high concentrated ownership and the flexibility of exchanging financial resources and managerial expertise within the group (Granovetter, 1995; Ghatak and Kali, 2001; Fisman and

Khanna, 2004; and Gopalan et al., 2007). Second, the period of our examination, which coincides with the adoption of market liberalization in early 1990s, ushered a period of significant activity in capital markets allowing us to examine the issue using a significantly large sample of equity offerings.

Using a sample of 1,290 Indian IPOs and 970 Indian SEOs during 1991-2012 period, we use two different approaches to analyze market timing of stand-alone and groups affiliated firms. First, following Alti (2006) we measure market timing by assessing whether the IPO or SEO takes place in a hot or a cold issue market. The rationale for this approach is that if issuers regard hot-issue markets as windows of opportunities then they should react by issuing more equity in hot compared to cold markets (Alti, 2006). By measuring market timing in this manner also helps us to avoid the confounding effects of firm-specific characteristics. We supplement this approach by using calendar-time method to measure the long-run performance of stand-alone and group-affiliated IPOs and SEOs. More specifically, we use Carhart's (1997) four-factor model to analyze the long-run performance which serves as our indirect test of market timing.

The main finding of our study is that there are significant differences between group-affiliated and stand-alone firms in how they access the external capital market. We find that stand-alone firms raise significantly larger proceeds through IPOs while group-affiliated firms significantly outweigh stand-alone firms in raising capital through SEOs. More importantly, while we find that stand-alone firms raise significantly larger proceeds through their IPOs in a hot-issue market, we fail to find any significant difference in equity proceeds raised by group-affiliated firms between hot and cold IPOs markets. Conversely, we find that group-affiliated firms raise significantly larger proceeds through SEOs in hot-issue markets. The difference in SEO proceeds raised by stand-alone firms between hot-issue and cold-issue markets is not significant.

We follow this up by examining the three-year long-run performance of IPOs and SEOs issued by stand-alone and group-affiliated firms in hot and cold periods. We find that among stand-alone firms, hot-issue IPOs under-perform significantly relative to cold-issue IPOs whereas there is no significant difference in the performance of hot-issue and cold-issue SEOs. On the other hand, among group-affiliated firms, hot-issue SEOs under-perform significantly to cold-issue SEOs whereas the long-run performance of hot-issue IPOs and cold-issue IPOs does not differ significantly.

Taken together, the results on the long-run performance and the amount of equity issuance in hot markets provide convincing evidence of the divergent behavior of stand-alone and group-affiliated firms in raising external equity. More specifically, we find that stand-alone firms appear to time the IPO market while group-affiliated firms appear to time the SEO market. Our results, for group-affiliated firms, are consistent with Masulis et al. (2016) who find that because of group certification these firms are able to raise funds even in weak IPO markets at favourable terms. As for stand-alone firms, without group certification and the presence of information asymmetry, they are more likely to access the markets when conditions are favorable. The findings also suggest that SEO is a more important fund raising vehicle for group-affiliated firms. Our study makes an important contribution to the existing literature. To the best of our knowledge, this is the first study to examine market timing for group-affiliated and stand-alone firms. We complement the previous literature (Gopalan et al., 2007; Almeida et al., 2015; Masulis et al., 2016) and show significant differences in how the two firm types access external capital market.

The remainder of the paper is organized as follows: Section 2 describes the data, discusses the methodology and provides summary statistics. Sections 4 and 5 discuss the empirical results. Section 6 makes some concluding remarks.

2 Data description, methodology and summary statistics

2.1 Data

Our sample period begins in 1991, the year that ushered financial deregulation and major economic reforms in India. We end the sample period in 2012 to allow us 3 years of long-run performance data after the offering. The IPO/SEO data is collected from Prime Database, a leading provider of IPO and other capital markets data in India. We supplement this with data from SDC database. We gather financial data from Prowess database which is a widely used source of Indian data in academic research (Marisetty and Subrahmanyam, 2010; Bubna and Prabhala, 2011; Gopalan and Gromley, 2013; Wadhwa et al., 2016). Our initial unfiltered sample contained 3,995 IPOs and 1,700 SEOs. From this, we excluded financial institutions and investment companies (699 IPOs and 108 SEOs); utilities (194 IPOs and 84 SEOs); privatized enterprises (103 IPOs and 34 SEOs). Further, following Lyol et al. (1999), we also eliminated 114 SEOs that were issued by same firm within three years of previous SEO offering. Lastly, we eliminated 1,709 IPOs and 390 SEOs from our sample due to non-availability of data. Hence, the total sample for this study comprises of 1,290 IPOs and 970 SEOs.

2.2 Methodology

In this section, we briefly describe our methodology and define hot and cold equity issue markets.

2.2.1 Measuring market timing

There are two primary approaches that previous literature has adopted in measuring market timing in equity issuance. On one hand, market timing has been examined using a forward looking approach in which the long-run performance of IPOs and SEOs is observed in 3 or 5 years in the post-equity issuance period (for example, Ritter, 1991; Loughran and Ritter, 1995; Brav et al., 2000). On the other hand, market timing has also been examined by looking at the relationship between the equity issuance activity of firm and the past market valuations or returns (for example, Pagano et al., 1998; Baker and Wurgler 2000; Hovakimian et al, 2001; and Baker and Wurgler, 2002) examine. In most of these studies, market-to-book ratio is considered as proxy of market valuation and positive relation of this ratio with the equity issuance is considered as the market timing by the issuers. In this paper, we use both the approaches in measuring market timing.

We primarily follow Alti's (2006) in examining the association between issuance activity and market valuation. For this, we divide the issuance activity into hot and cold periods and examine whether the likelihood as well as the amount of equity issuance is greater in hot periods. The advantage in following Alti's (2006) approach is that it allows us to isolate market timing and measure it as a function of market conditions without associating it with firm level characteristics. Moreover, this approach helps us in addressing two dimensions of market timing: (i) firms are more likely to go public when the issuers perceive that the market conditions are favorable and (ii) firms that go public at the time of favorable market conditions are likely to sell more equity than if they had gone public at the time when market conditions were unfavorable. In the forward-looking approach, we examine the long-run performance of IPOs and SEOs for 3 years after the equity issuance. We use calendar-time regression using Carhart's (1997) four factor model to examine the long-run performance. In addition to this, we also compare and contrast the difference between the long-run performance of hot and cold issues for stand-alone and group-affiliated offerings.

2.2.2 Definition of hot and cold markets

Since we measure market timing by examining the issuance of equity in hot and cold markets, in this subsection we describe how we define these two markets. Following Helwega and Liang (2004) and Alti (2006), we define hot and cold IPO/SEO markets on the basis of monthly IPO/SEO volume and IPO/SEO underpricing. First, we consider the number of IPOs/SEOs in each month between 1991 and 2012. Then, we compute the three month centered-moving average of number of IPOs/SEOs for each month. Since the Indian economy grew at the approximate rate of 6.6 % per annum during the 21-year time period, we detrend the three month centered moving average IPO/SEO volume at the rate of 0.55 % per month. We define hot months as those months that are above the median of the distribution of the detrended monthly moving average IPO/SEO volume and cold months as those that are below the median of the detrended monthly moving average IPO/SEO volume across all the months. We follow the same procedure of three months moving average to classify hot and cold markets on the basis of underpricing without detrending. Hot (cold) months are those in which underpricing is above (below) the three-month centered moving average of underpricing.

2.3 Summary Statistics

Table 1 provides the summary statistics of all our main variables. Appendix A provides the definitions of all the variables used in the study. Panel A reports summary statistics of the variables for all the equity issues (2260) i.e. total of IPOs and SEOs. The average proceeds to total assets (P/A) is 0.200, which suggests that the amount raised through equity issuance averages about 20% of total assets. On average, the shares issued during an offering is 31.7% of the total outstanding shares. The average offer/book is 93.43 and the average M/B ratio that is 1.5. Panel B shows summary statistics of our variables for IPOs and SEOs separately. Overall, the average proceeds to total assets for IPOs and SEOs is 0.25 and 0.15 respectively. The average percentage of shares issued by IPO firm (SEO firm) is 37.48% (23.99%) of the total outstanding shares. Just as for the overall sample, the average offer to book ratio for IPOs is 82.73 and 107.65 for SEOs, which means that the firms sell their equity at a price which is higher than the book value. This evidence is further substantiated by the average M/B which is 1.55 for IPOs and 1.42 for SEOs.

Panel C reports the summary statistics for stand-alone firms and shows that the average proceeds to assets for IPOs is 0.25 and 0.13 for SEOs. The average shares issued by stand-alone firms at IPO is 39.06% and at SEOs is 25.86%. Further, while the average deal size for IPOs is 662.31 million rupees, it is only 322.91 million rupees for SEOs. Overall, it appears that stand-alone firms issue more equity in IPOs relative to the amount raised in SEOs.[1] In terms of market timing variables, both offer to book ratio and M/B is high for IPOs and SEOs suggesting a tendency of stand-alone firms to sell overvalued equity. Panel D reports the summary statistics for group-affiliated firms. In contrast to stand-alone firms, it appears that group-affiliated firms raise more equity from SEOs than IPOs. The average equity proceeds from SEOs is 3,885.83 million rupees while it is only 1959.77 million rupees from IPOs. The offer to book ratio is also higher for group-affiliated SEOs (123) compared to group-affiliated IPOs (103.83). Just as in other equity issues, the average M/B of group-affiliated IPOs and SEOs is also greater than 1.

[Please insert table 1 about here]

3 Hot markets and market timing

3.1 Univariate analysis

In this section, we perform a univariate analysis of average equity proceeds (both IPO and SEO) issued in hot versus cold markets by stand-alone and group-affiliated firms. Table 2 reports

[1] We provide further evidence to this in results of univariate analysis reported in Table 2

the results of our univariate analysis to examine the difference in average equity proceeds raised in hot and cold issue markets. All the variables are defined in Appendix A. The different proxies of proceeds are: P/A_t, P/A_{t-1}, %Issued, Offer/Book and LnDSize. Proceeds to Total Assets (P/A_t) which is defined as the total IPO/SEO proceeds divided by IPO/SEO year-end total assets; Proceeds to lag of Total Assets (P/A_{t-1}) which is defined as the total IPO/SEO proceeds divided by total assets at the beginning of the IPO/SEO year; %Issued is defined as number of shares issued in IPO/SEO divided by shares outstanding at the year-end; Offer/Book is ratio of offer price at which IPO/SEO share is offered to the book value of the share; and LnDSize is the natural log of equity proceeds in millions from IPO/SEO where proceeds is computed as offer price multiplied by number of shares issued in an offering. Columns (1) – (5) report the univariate analysis for stand-alone firms whereas columns (6) – (10) reports the same analysis for group-affiliated firms. Panel A shows that stand-alone firms issue substantially more equity through IPOs in hot-issue market compared to cold-issue market as the difference between the proceeds of hot and cold-issue market is positive and significant in almost all the cases. For example, the proceeds from sale of equity are 25% of the year-end total assets for hot-issue market firms where that same ratio for cold-issue market firms is only 22%. Likewise, the percentage of shares issued in hot-issue market is 36.39% of the total year-end outstanding shares compared to 30.53% for cold-issue market. The other variables capturing equity proceeds like offer-to-book ratio and gross equity proceeds from the sale of share are also higher. This market timing effect is significant and robust across both the classifications of hot and cold markets – monthly volume and underpricing. However, there is no significant difference in the SEO equity proceeds between hot and cold markets for stand-alone firms.

Interestingly, however, the results in columns (6) – (10) show that group-affiliated firms sell substantially more equity through SEOs in the hot-issue market compared to SEOs issued in the cold-issue market. Most of the differences between proceeds between hot and cold-issue markets are positive and significant. For example, when hot and cold issue market firms are classified on the basis of SEO monthly volume, the proceeds from sale of equity are 34% of the year-end total assets for hot-issue market firms where that same ratio for cold-issue market firms is only 11%. Likewise, the offer to book ratio in hot-issue market is 209.20 compared to 93.40 only for cold-issue market. The other variables capturing equity proceeds like P/A_{t-1} is also higher for hot as compared to cold issue market firms. This market timing effect is significant and robust across both the classifications of hot and cold markets – monthly volume and underpricing. On the other hand, the difference between the average proceeds of hot and cold group-affiliated firms IPOs is not significant. Hence, our univariate analysis shows that among stand-alone firms, IPOs issue more equity in hot markets as compared to cold markets whereas among group-affiliated firms, SEOs raise more equity proceeds from hot markets than the cold markets.

[Please insert table 2 about here]

3.2 Multivariate Evidence

In this section, we conduct a multivariate regression analysis to assess market timing of stand-alone and group-affiliated firms. The evidence of hot-market effect on the equity proceeds shown in univariate analysis section could be due to different characteristics of hot and cold-issue market firms (Alti, 2006). To address this issue, we run the following regression which controls for various other determinants of equity issues:

$$Y_t = \propto + \beta_1 HOT + \beta_2 \frac{M}{B}_t + \beta_3 \frac{EBITDA}{A}_{t-1} + \beta_4 Lsize_{t-1} + \beta_5 \frac{PPE}{A}_{t-1} + \beta_6 \frac{D}{A}_{t-1} + D_{Ind} + \varepsilon_t \quad \text{.....(1)}$$

In the above equation, Y_t takes one of the five different variables representing equity issuance: P/A_t, P/A_{t-1}, %Issued, Offer/Book and LnDSize. The time subscript *t* represents the IPO year. Intercept of the regression captures average of dependent variable of cold IPO firms. Our main variable of interest is the 'HOT' dummy variable which takes the value of 1 if the firm issues equity in the hot months and 0 otherwise. The rationale for considering the 'HOT' dummy variable is that if the issuers regard hot-issue markets as the windows of opportunity then they should react by issuing more equity than they would do otherwise (Alti, 2006). Hence, if β_1 is positive and significantly different from zero then it reflects the market timing attempts by issuers. We also include several other control variables that are considered as determinants of equity issue by previous literature (Rajan and Zingales, 1995 and Alti, 2006): market-to-book ratio, profitability ratio, firm size, tangibility and lagged leverage. We estimate the above cross-sectional regression with industry fixed effects. All the variables are defined in Appendix A. We run the above regression equation for four different categories of firms: stand-alone IPOs, stand-alone SEOs, group-affiliated IPOs and group-affiliated SEOs.

Table 3 reports the regression results to examine the market timing effects on the IPO issuance. Panel A shows the results when HOT is defined on the basis of monthly IPO volume, while B shows the results when HOT is defined on the basis of IPO underpricing. In each of these panels, the regression results for stand-alone firms are reported in columns (1) – (5) and for group-affiliated IPOs in columns (6) – (10).

The results in Panel A in which 'hot' and 'cold' period classification is based on the monthly IPO volume shows that the 'HOT' variable for stand-alone IPO firms is positive and significant for four out of five proxies of proceeds. However, consistent with univariate analysis, we find that the 'HOT' variable is not significant for group-affiliated IPOs. The results are almost similar in Panel B, where hot and cold IPOs are classified on the basis of underpricing. Overall, the results suggest that equity proceeds raised by hot-issue stand-alone IPOs are substantially more than the equity proceeds of cold-issue stand-alone IPOs. For example, Panel A shows that P/A_t of hot stand-alone IPOs is 0.05 higher than the cold stand-alone IPOs. Likewise, P/A_{t-1} of cold stand-alone IPOs is 0.51 where for hot stand-alone IPOs, the ratio is 0.70 (0.51 + 0.19), Offer/Book ratio is also higher i.e. 51.8 for hot stand-alone firms and lower i.e. 42.08 for cold stand-alone IPOs. Finally, the deal size of hot stand-alone IPOs is 0.39 higher than the cold stand-alone IPOs. This indicates genuine market timing effect in case of stand-alone IPOs, however, no such effect is seen in case of group-affiliated IPOs as there is no difference in the equity proceeds of hot-issue and cold-issue group-affiliated IPOs.

The other determinants of equity issuance activity such as profitability, size, leverage, etc., show reasonable significance. For example, when the firms go for equity issuance, they tend to decline their debt levels which results in low leverage. Literature (see Baker and Wurgler, 2002; Elliot et al., 2008) also shows that during favorable market conditions, firms issue equity and retire debt whereas during unfavorable market conditions, firms issue debt. Thus, we expect a negative relation between equity issuance and debt to equity ratio. Table 3 also shows such relationship: beta coefficients of D/A_{t-1} are negative in all the cases. In case of stand-alone firms, small and more profitable firms go public to raise external equity for further expansion. It is difficult for less profitable stand-alone firms to go public because such firms do not attract public confidence due to low profitability. On the other hand, in case of business groups, small and less profitable firms go public because profitable firms can rely on the their internal resources and it is the less profitable firms that require external financing for further expansion and this can be easily done with group support. Lastly, in most of the cases, market-to-book ratio is found to have a positive relation with equity issuance which shows that growth firms tend to issue more equity. Hence, the regression results confirm our univariate findings and show that stand-alone firms time the market while going for IPOs by issuing more equity in hot-issue months as compared to cold-issue months.

[Please insert table 3 about here]

Next, we examine the market timing effects on the issuance of SEOs. Table 4 reports the regression results of equation (1) to examine the market timing effects on the equity proceeds of stand-alone SEOs. Panel A and Panel B report the regression results when the hot and cold issue market is classified on the basis of monthly SEO volume and underpricing respectively. Both Panel A and B show that the 'HOT' variable is not significant in case of stand-alone SEOs whereas the variable is positive and significant in group-affiliated SEOs. For example, Panel A shows that P/A_t of hot group-affiliated SEOs is 0.21 higher than the cold group-affiliated SEOs. Likewise, P/A_{t-1} of cold group-affiliated SEOs is 2.10 where for hot group-affiliated SEOs, the ratio is 2.62 (2.10 + 0.52), Offer/Book ratio is also higher i.e. 259.23 (150.44 + 108.79) for hot group-affiliated SEOs and lower i.e. 150.44 for group-affiliated SEOs. Finally, the deal size of hot group-affiliated SEOs is 0.43 higher than the cold group-affiliated SEOs. This indicates genuine market timing effect of case of group-affiliated SEOs. The other determinants of equity issuance activity for stand-alone and group-affiliated SEOs are also found to have reasonable significance. For example, in case of stand-alone firms, small and more profitable firms issue equity for further expansion. It is difficult for less profitable stand-alone firms to issue follow-up equity as such firms do not attract public confidence due to low profitability. On the other hand, in case of business groups, small and less profitable firms issue follow-up equity because profitable firms can rely on the their internal resources and it is the less profitable firms that require external financing for further expansion and this can be easily done with group support. Lastly, in most of the cases, market-to-book ratio is found to have a positive relation with equity issuance which shows that growth firms tend to issue more equity. We find the similar results when we classify hot and cold markets on the basis of underpricing, the results of which are shown in Panel B of the table. Thus, these findings show that stand-alone firms do not time the market while issuing equity through SEOs whereas among group-affiliated firms, hot-issue SEOs issue substantially more equity as compared to cold-issue SEOs. Hence, we can say that group-affiliated firms time the market while issuing equity through SEOs.

[Please insert table 4 about here]

From the above results, we can say that our analyses provide a clear evidence of the market timing in India by stand-alone IPOs and group-affiliated SEOs. Hot-issue market stand-alone IPOs sell substantially more equity than their cold-market counterparts. On the other hand, among group-affiliated firms, hot-issue market SEOs sell substantially more equity as compared to cold-issue markets SEOs. Stand-alone SEOs and group-affiliated SEOs do not time the market and exhibit any opportunistic behavior against outside investors.

4 Long-run performance

In this section, we further analyze market timing by examining the long-run performance of stand-alone and group-affiliated IPOs and SEOs. The long-run underperformance of IPOs and SEOs which is well-documented in the literature (Ritter, 1991; Loghran and Ritter, 1995, 1997;

Kim and Weisbach, 2008; Hertzel and Li, 2010) is attributed to market timing exercised by equity issuing firms. Following Hertzel and Li (2010) and Wadhwa et al. (2016), we use calendar-time approach to examine the long-run performance of IPOs and SEOs for three years after the equity issuance. To examine the long-run performance of firms after equity issuance, we use Carhart's (1997) four factor model. We also report our performance results for other two simpler versions of Carhart (1997) model that are: capital assets pricing model (CAPM) and Fama- French (1993) three-factor model. Carhart (1997) four factor model is given as:

$$R_{pt} - R_{ft} = \alpha + \beta_1(R_{mt} - R_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \varepsilon_t \ldots\ldots\ldots\ldots(2)$$

In the above regression equation, R_{pt} is the monthly portfolio returns of the IPOs/SEOs calculated for the month t. Intercept α is a measure of long-run performance. If α is significantly different from zero then it is an indication of abnormal long-performance of the firm in the post-equity issuance period of the firm. Positive α denotes over-performance and negative α denotes under-performance. We follow the standard procedure for the construction of factors: SMB_t, HML_t and MOM_t as given by Fama and French (1993) and Carhart (1997). We also examine the difference in the long-run performance between issues made in hot and cold markets by using the following model:

$$H_{pt} - C_{pt} = \alpha + \beta\,(R_{mt} - R_{ft}) + s\,SMB_t + h\,HML_t + \varepsilon_t \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(3)$$

In the above equation, α denotes the difference in the long-run performance of issues between hot and cold markets. All the variables of the equations (2) and (3) are defined in Appendix A. Negative α indicates that hot-issue firms show greater underperformance than cold-issue firms and positive α denotes vice versa. Greater underperformance of hot-issue firms than that of cold-issue firms further suggests that hot-issue firms time the market while issuing equity.

Table 5 reports the results of calendar-time regression based on equation (2) for IPOs which examines the three-year long-run performance when hot and cold IPOs are classified on the basis of monthly IPO volume.[2] Columns (1) – (3) show the long-run performance of hot and columns (4) – (6) show the performance of cold market issues. Columns (7) – (9) show the difference in the long-run performance between hot and cold IPOs using equation (3). α in columns (1) – (3) for is negative in the all the models which suggests underperformance of stand-alone IPOs in the hot market in the post listing period. Although, stand-alone IPOs issued in the cold market also show negative long-run underperformance, but α in columns (4) – (6) are not statistically significant. More importantly, however, the difference in the long run performance between stand-alone IPOs issued in and hot and cold market is both economically and statistically significant. For instance, using the CAPM model, we find that an investor's underperformance on account of investment in stand-alone IPOs issued in the hot market is about 0.04 % per month (1.44 % on a three year basis) relative to the investment in IPOs issued in the cold market.

Results of calendar time regressions examining the long-run performance of group-affiliated IPOs issued in hot and cold markets are reported in Panel B of Table 5. Interestingly, however consistent with our earlier findings, we find that in relative terms groups affiliated IPOs issued in the cold market tend to underperform more than those issued in the hot market. For example, the intercept terms in columns (4) – (6) for cold group-affiliated IPOs are negative and significant whereas the intercept terms for hot group-affiliated IPOs are not significant in two out of three model models. It is negative only in CAPM model and that too it is smaller i.e. 0.04% (1.44 % on a three year basis) compared to the intercept of cold group-affiliated IPOs i.e. 0.08%

[2] Due to space constraints, we do not report the calendar-time regression results when hot and cold issue markets are classified on the basis of underpricing. The underpricing regression results are identical to the reported regression results and are available upon request.

per month (2.88 % on a three year basis). The intercept terms in columns (7) – (9) denoted by α is, however, not significant. This finding of ours is also consistent with earlier univariate and multivariate analysis of group-affiliated IPOs that show that there is no significant difference between the equity proceeds raised from hot and cold group-affiliated IPOs. Hence, we conclude that group-affiliated firms do not appear to time the market while going public.

[Please insert table 5 about here]

We next focus on the performance of SEOs the results of which is reported in Table 6. Panel A reports the long-run performance and difference in long-run performance of hot and cold stand-alone SEOs. As in Table 5, columns (1) – (3) show the long-run performance of hot and columns (4) – (6) show the performance of cold market issues. Further, columns (7) – (9) show the difference in the long-run performance between hot and cold SEOs. Although columns (2) and (3) show that stand-alone SEOs issued in the hot market appear to underperform, overall we fail to find any difference in post issuance performance between stand-alone SEOs issued in the hot and cold markets. We rerun this analysis for group-affiliated SEOs and present the results in Panel B of Table 6. Columns (1) – (3) show that group-affiliated SEOs issued in the hot market underperform in the post issuance period with decline in stock returns of almost -0.05 % (using CAPM model) per month (or -1.9% after 3 years). More importantly, we find that the difference in long-run performance between hot and cold SEOs is negative and significant (columns (7) – (9)). Using the CAPM model we find that an investor loss would be around 0.08% per month (or 2.9 % in three years) more from the investment in SEO issued in the hot market relative to SEO issued in the cold market. Hence, through this, we provide the evidence that the group-affiliated SEOs time the market by selling overvalued equity in hot issue market. Again, this result is also consistent with our univariate and multivariate analyses of group-affiliated SEOs where we show that hot group-affiliated SEOs issue more equity than the cold group-affiliated SEOs.

[Please insert table 6 about here]

From the above analyses, we conclude that in India, stand-alone firms time the market while issuing equity for the first time. On the other hand, group-affiliated firms time the market while issuing equity during follow-up offerings.

5 Conclusions

Market timing in equity raising has been a topic that has caught attention of many researchers. While the issue has been examined in some detail for the developed markets, evidence is relatively limited for emerging markets. Among others, one of the unique features of the emerging markets is the presence of group-affiliated firms. Owing to the presence of internal capital markets for group-affiliated firms, we argue that these firms should behave differently than stand-alone firms in accessing external equity markets. We examine this issue in the context of the Indian equity market, which provides a suitable setting as Indian business landscape is dominated by group-affiliated firms. Our sample comprises of 1290 IPOs and 970 SEOs issued over a 21-year period from 1991 to 2012. Of these, there are 1050 IPOs and 516 SEOs undertaken by stand-alone firms and 240 IPOs and 454 SEOs issued by group-affiliated firms.

In the main results, we find strong evidence that stand-alone firms issue more equity through IPO in the hot-markets relative to those issued in cold markets. We fail to find similar results of SEOs issued by stand-alone firms. These results are robust to our long-run performance analysis. Hot-issue stand-alone IPOs show greater long-run underperformance than cold-issue IPOs suggesting that stand-alone firms appear to time the market in issuing their IPOs. We find quite different results for Group-affiliated firms. Group-affiliated firms appear to issue more equity in hot-issue relative to cold-issue market when raising capital through SEOs. On the other hand, there is no difference between the equity proceeds of hot-issue and cold-issue markets for IPOs. We find that results are similar when we examine the post-issuance long run performance. Overall, we conclude that group-affiliated firms appear to time when issuing SEOs and stand-alone firms appear to do the same when issuing IPOs.

Appendix A. Description of variables used in the study

Variables	Definition
P/A	P/A is measured as IPO/SEO proceeds divided by IPO/SEO year-end total assets where proceeds are computed as offer price multiplied by no. of shares issued in an offering.
%Issued	%Issued is computed as number of shares issued in IPO/SEO divided by shares outstanding the year-end.
Offer/Book	Offer/Book is ratio of offer price at which IPO/SEO share is offered to the book value of the share.
DSize	DSize_million is the equity proceeds in millions from IPO/SEO where proceeds is computed as offer price multiplied by no. of shares issued in an offering.
M/B	M/B is the market-to-book ratio of IPO/SEO at the year-end.
EBITDA/A	EBITDA/A is measured as Earnings before Interest, Tax, Depreciation and Amortization to year-end assets.
Size	Size is year-end net sales of the firm.
PPE/A	PPE/A is measured as net plant, property, and equipment divided by year-end total assets.
D/A	D/A is debt to year-end total assets.
Rm	Rm is the market return. Return on NSE CNX 500 is used as a proxy of market return.
Rf	Rf is the risk-free rate. Interest on monthly 91 day T-bill is used as proxy of risk-free rate.
SML	Return on the portfolio of small cap firms minus return on the portfolio of large cap firms.
HML	Return on the portfolio of high book-to-market firms minus return on the portfolio of low book-to-market firms.
Mom	Return on the portfolio of past winner firms minus return on the portfolio of low winners.
Hpt	Return on the portfolio of hot-issue market firms
Cpt	Return on the portfolio of cold-issue market firms
Hot IPO/SEO	An IPO/SEO is considered as hot IPO/SEO if the 3 months moving average IPO/SEO volume/underpricing is more than the median 3 months moving average IPO/SEO volume/underpricing.
Cold IPO/SEO	An IPO/SEO is considered as hot IPO/SEO if the 3 months moving average IPO/SEO volume/underpricing is less than the median 3 months moving average IPO/SEO volume/underpricing.

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Table 1: Descriptive Statistics of all the variables

	Mean	SD	Min	Median	Max
Panel A:	All Equity Issues (N=2260)				
P/A	0.200	0.430	0.000	0.147	16.210
%Issued	31.690	117.520	0.001	25.088	5186.900
Offer/Book	93.430	183.340	0.298	58.894	4666.740
DSize_million	1290.00	5176.18	0.004	62.18	91875.00
M/B	1.500	3.180	-76.835	0.980	69.510
EBITDA/A	0.120	0.080	-0.356	0.114	1.500
PPE/A	0.200	0.160	0.000	0.162	0.990
D/A	0.460	0.260	0.001	0.451	1.840

	Mean	SD	Min	Median	Max	Mean	SD	Min	Median	Max
Panel B:	IPOs (N=1290)					SEOs (N=970)				
P/A	0.247	0.270	0.003	0.203	7.110	0.150	0.580	0.000	0.056	16.210
%Issued	37.482	145.330	0.587	28.537	5186.900	23.990	63.250	0.001	12.331	1644.700
Offer/Book	82.734	80.310	1.020	64.396	1171.450	107.650	263.490	0.298	46.896	4666.740
DSize_million	627.51	3618.24	2.52	34.97	91875.00	2171.05	6609.46	0.004	154.37	70929.00
M/B	1.558	2.770	0.003	1.006	69.510	1.420	3.650	-76.835	0.935	19.950
EBITDA/A	0.125	0.080	-0.171	0.120	0.600	0.110	0.090	-0.356	0.103	1.500
PPE/A	0.178	0.150	0.000	0.148	0.990	0.220	0.170	0.000	0.187	0.930
D/A	0.386	0.200	0.001	0.388	1.220	0.550	0.280	0.001	0.551	1.840
Panel C:	SA IPOs (N=1050)					SA SEOs (N=516)				
P/A	0.248	0.280	0.003	0.207	7.110	0.126	0.190	0.000	0.060	1.990
%Issued	39.066	160.480	0.587	29.644	5186.900	25.862	77.470	0.004	13.481	1644.700
Offer/Book	77.911	66.850	1.020	64.661	1171.450	94.144	167.570	0.298	47.256	1617.250
DSize_million	662.31	2678.45	0.004	94.01	38958.06	322.99	1059.30	2.52	32.17	18993.48
M/B	1.482	2.840	0.003	0.988	69.510	1.356	1.920	-8.581	0.876	15.470
EBITDA/A	0.121	0.080	-0.167	0.117	0.600	0.099	0.080	-0.314	0.094	0.520

PPE/A	0.177	0.140	0.001	0.150	0.990	0.199	0.160	0.000	0.169	0.910
D/A	0.380	0.200	0.001	0.382	1.220	0.538	0.310	0.001	0.545	1.420

Panel D:	BG IPOs (N=240)					BG SEOs (N=454)				
P/A	0.240	0.250	0.005	0.171	2.770	0.180	0.820	0.000	0.047	16.210
%Issued	30.550	28.450	4.101	25.203	411.500	21.860	41.540	0.001	10.893	491.380
Offer/Book	103.830	120.920	1.995	61.941	853.180	123.000	340.800	0.363	46.045	4666.740
DSize_million	1959.77	7968.22	9.00	70.19	91875.00	3885.83	8930.25	0.041	443.73	70929.00
M/B	1.890	2.410	0.028	1.122	19.680	1.490	4.930	-76.835	1.018	19.950
EBITDA/A	0.140	0.080	-0.171	0.140	0.500	0.120	0.100	-0.356	0.111	1.500
PPE/A	0.180	0.160	0.000	0.137	0.860	0.240	0.180	0.000	0.213	0.930
D/A	0.410	0.200	0.006	0.422	0.870	0.570	0.250	0.001	0.557	1.840

Notes: This table reports the descriptive statistics of all the variables. *Panel A* reports the summary statistics for all equity issues - Indian IPOs and SEOs together. IPO dataset includes 1290 IPOs listed on Bombay Stock Exchange (BSE) and/or National Stock Exchange (NSE) between 1991 and 2012. SEO dataset includes 970 SEOs that issued equity in India between 1991 and 2012. P/A is measured as IPO/SEO proceeds divided by IPO/SEO year-end total assets. %Issued is computed as number of shares issued in IPO/SEO divided by shares outstanding the year-end. Offer/Book is ratio of offer price at which IPO/SEO share is offered to the book value of the share. DSize_million is the equity proceeds in millions from IPO/SEO. M/B is the market-to-book ratio of IPO/SEO at the year-end. EBITDA/A is measured as Earnings before Interest, Tax, Depreciation and Amortization to year-end assets. PPE/A is measured as net plant, property, and equipment divided by year-end total assets. D/A is debt to year-end total assets. *Panel B* shows the summary statistics of total IPOs and SEOs. *Panel C* reports the summary statistics of stand-alone IPOs and SEOs whereas *Panel D* reports the summary statistics of IPOs and SEOs affiliated with business group firms.

Table 2: Univariate Analysis

	Stand-alone Firms					Group-affiliated Firms				
	P/A_t (1)	P/A_{t-1} (2)	%Issued (3)	Offer/Book (4)	LnDSize (5)	P/A_t (6)	P/A_{t-1} (7)	%Issued (8)	Offer/Book (9)	LnDSize (10)
IPOs (Monthly Volume)										
Hot	0.25	0.56	36.39	77.63	18.01	0.24	0.73	30.14	104.30	18.90
Cold	0.22	0.42	30.53	72.53	17.43	0.25	0.66	31.83	103.30	18.32
Diff	0.04***	0.14**	5.86*	5.10	0.58***	0.01	0.07	1.69	0.97	0.58
t-value	2.46	2.15	1.84	0.89	4.26	0.34	0.21	0.4	0.05	1.85
IPOs (Underpricing)										
Hot	0.26	0.60	48.34	85.41	18.15	0.22	0.62	23.78	93.63	18.48
Cold	0.24	0.48	31.09	69.28	17.65	0.29	0.95	33.17	131.40	19.48
Diff	0.02	0.11***	17.25*	16.13***	0.50***	0.06	0.33	-9.39	37.75	1.00
t-value	1.3	2.47	1.76	3.93	5.16	1.68	1.05	-2.3	2.17	3.28
SEOs (Monthly Volume)										
Hot	0.12	0.23	22.48	86.77	18.44	0.34	0.75	21.46	209.20	20.16
Cold	0.14	0.36	34.71	106.90	18.69	0.11	0.19	20.73	93.40	19.98
Diff	-0.01	-0.13	-12.24	-20.17	-0.24	0.23***	0.56**	0.73	115.80***	0.18
t-value	-0.84	-1.32	-1.55	-1.23	-1.33	2.64	2.18	0.19	3.19	0.72
SEOs (Underpricing)										
Hot	0.12	0.21	32.59	97.16	18.35	0.19	0.33	22.11	172.80	20.59
Cold	0.13	0.31	21.70	92.06	18.58	0.12	0.22	19.88	86.73	19.53
Diff	-0.01	-0.10	10.89	5.10	-0.23	0.06*	0.11	2.23	86.06***	1.06***
t-value	-0.47	-1.12	1.6	0.34	-1.36	1.80	1.16	0.64	2.58	4.77

Notes: This table reports the univariate analysis of difference between equity proceeds variables of hot and cold markets of stand-alone and group-affiliated firms. Hot and cold market firms are classified on the basis of monthly IPO/SEO volume and IPO/SEO underpricing. Columns (1)-(5) show the difference in mean test of proceeds for stand-alone firms and columns (6)-(10) show the difference in mean test of proceeds for group-affiliated firms. All the variables are defined in Table 1. *** statistically significant at 1%

** statistically significant at 5%

* statistically significant at 10%

Table 3: Market Timing Effects on IPO Issuance Activity

Panel A: Monthly Volume

	Stand-alone Firms					Group-affiliated Firms				
	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Intercept	0.25***	0.51***	29.89	42.08***	14.93***	0.19***	1.25**	48.86***	24.43	14.92***
	(7.22)	(5.74)	(1.45)	(4.84)	(106.19)	(2.86)	(2.19)	(6.43)	(0.83)	(34.25)
Hot	0.05**	0.19***	14.91	9.72*	0.39***	0.00	0.24	-0.24	2.36	0.30
	(2.24)	(3.12)	(1.04)	(1.77)	(4.04)	(-0.03)	(0.74)	(-0.06)	(0.14)	(1.24)
M/B_t	0.00	0.01	-1.12	1.95***	0.00	0.02***	0.09	-0.71	12.45***	0.15***
	(-0.37)	(0.99)	(-0.64)	(2.90)	(0.19)	(2.34)	(1.44)	(-0.91)	(4.08)	(3.37)
$EBITDA/A_{t-1}$	0.27***	2.80***	9.65	113.64***	0.50	0.21*	1.25	-1.32	114.81***	-0.58
	(2.87)	(11.72)	(0.17)	(5.39)	(1.32)	(1.92)	(1.35)	(-0.11)	(2.40)	(-0.82)
$Lsize_{t-1}$	-0.01*	-0.10***	-2.65	4.59***	0.51***	-0.01	-0.17**	-2.34***	5.21	0.53***
	(-1.84)	(-8.46)	(-0.98)	(4.42)	(27.55)	(-0.99)	-(2.40)	(-2.48)	(1.42)	(9.86)
PPE/A_{t-1}	-0.04	-0.16	26.44	-19.71**	0.21	0.02	0.34	5.32	-24.52	-0.32
	(-0.84)	(-1.43)	(1.00)	(-1.95)	(1.18)	(0.28)	(0.51)	(0.60)	(-0.71)	(-0.63)
D/A_{t-1}	-0.11***	-0.23**	-21.24	-52.49***	-0.55***	-0.03	-1.38**	0.28	-59.46*	-0.81*
	(-2.62)	(-2.21)	(-0.87)	(-5.47)	(-3.29)	(-0.46)	(-2.19)	(0.03)	(-1.83)	(-1.70)
Ind. Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.03	0.17	0.01	0.15	0.53	0.06	0.09	0.06	0.21	0.45
Adj. R^2	0.02	0.16	0.00	0.14	0.53	0.03	0.07	0.03	0.19	0.43
N	1044	1044	1044	1046	1044	239	239	239	239	239

Panel B: Underpricing

	Stand-alone Firms					Group-affiliated Firms				
	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Intercept	0.29***	0.63***	33.13*	66.97***	15.30***	0.24***	1.68***	43.06***	42.37	15.40***
	(9.22)	(8.02)	(1.83)	(9.55)	(123.09)	(3.45)	(2.82)	(5.44)	(1.37)	(33.79)
Hot	0.02	0.06	19.51*	11.64***	0.19***	-0.06	-0.33	6.58	-19.10	-0.33
	(1.34)	(1.26)	(1.87)	(2.88)	(2.63)	(-1.57)	(-1.03)	(1.57)	(-1.17)	(-1.35)
M/B_t	0.00	0.01	-1.59	2.21***	0.01	0.02**	0.08	-0.72	12.48***	0.15***
	(-0.60)	(0.75)	(-0.90)	(3.23)	(0.45)	(2.37)	(1.43)	(-0.93)	(4.10)	(3.34)
$EBITDA/A_{t-1}$	0.26***	2.78***	-2.80	125.91***	0.70*	0.21**	1.19	-1.39	114.51***	-0.66
	(2.74)	(11.53)	(-0.05)	(5.82)	(1.82)	(1.95)	(1.29)	(-0.11)	(2.42)	(-0.94)
$Lsize_{t-1}$	-0.01	-0.09***	-1.39	3.88***	0.50***	-0.01	-0.18***	-2.03**	4.36	0.52***
	(-1.43)	(-7.86)	(-0.50)	(3.60)	(26.16)	-(1.31)	(-2.50)	(-2.13)	(1.17)	(9.54)
PPE/A_{t-1}	-0.04	-0.18	23.88	-20.06**	0.21	0.03	0.35	4.36	-21.98	-0.32
	(-0.97)	(-1.56)	(0.91)	(-1.97)	(1.15)	(0.40)	(0.52)	(0.49)	(-0.64)	(-0.62)
D/A_{t-1}	-0.11***	-0.21**	-26.84	-55.61***	-0.37**	-0.02	-1.27**	-1.53	-54.07*	-0.70
	(-2.57)	(-2.00)	(-1.09)	(-5.82)	(-2.16)	-(0.25)	(-2.00)	(-0.18)	(-1.65)	(-1.45)
Ind. Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.02	0.16	0.01	0.14	0.53	0.07	0.10	0.07	0.21	0.45
Adj. R^2	0.02	0.16	0.00	0.14	0.53	0.04	0.07	0.04	0.19	0.43
N	1048	1048	1048	1048	1048	239	239	239	239	239

Notes: This table reports regression results of the market timing effects on the IPO issuance activity of stand-alone and group-affiliated firms using the following regression equation:

$$Y_t = \alpha + \beta_1 HOT + \beta_2 M/B_t + \beta_3 EBITDA/A_{t-1} + \beta_4 Lsize_{t-1} + \beta_5 PPE/A_{t-1} + \beta_6 D/A_{t-1} + D_{Ind} + \varepsilon_t$$

where Y_t equals P/A_t, P/A_{t-1}, %Issued, Offer/Book and LnDSize. t subscript denotes the IPO year. The dummy variable HOT takes the value 1 if the firm issued equity in the hot-issue market and 0 otherwise. Lsize is the log of net sales in millions. D_{Ind} is industry dummy. All the regressions are estimated with industry-fixed effects. All other variables are defined in Table 1. Panel A reports the difference of mean test and regression results when

the firms are classified into hot and cold-issue market on the basis of monthly IPO volume and Panel B reports these results when the firms are classified into hot and cold-issue market on the basis of underpricing. *t*-statistics are given in the parentheses and have been estimated using clustered standard errors.
*** statistically significant at 1%
** statistically significant at 5%
* statistically significant at 10%

Table 4: **Market Timing Effects on SEO Issuance Activity**

Panel A: Monthly Volume

	Stand-alone Firms					Group-affiliated Firms				
	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Intercept	0.22***	1.00***	6.95	33.75	14.30***	0.85***	2.10***	49.94***	150.44	15.69***
	(5.73)	(4.56)	(0.36)	(0.90)	(38.32)	(3.64)	(3.10)	(4.92)	(1.57)	(29.39)
Hot	-0.02	-0.09	-11.17	-23.58	-0.16	0.21***	0.52**	0.87	108.79***	0.43**
	(-1.18)	(-1.04)	(-1.38)	(-1.51)	(-1.00)	(2.50)	(2.07)	(0.23)	(3.08)	(2.20)
Mt/B_t	0.01***	0.02	-1.65	23.58***	0.11***	0.01	0.01	-0.01	16.75***	0.04**
	(3.54)	(1.10)	(-0.82)	(6.09)	(2.97)	(0.72)	(0.58)	(-0.03)	(5.07)	(2.15)
$EBITDA/A_{t-1}$	0.14*	1.51***	10.00	190.05***	0.08	-1.03***	-3.11***	-4.66	143.50	-0.97
	(1.85)	(3.50)	(0.26)	(2.58)	(0.11)	(-3.37)	(-3.50)	(-0.35)	(1.14)	(-1.39)
$Lsize_{t-1}$	-0.02***	-0.21***	-0.98	2.73	0.48***	-0.07***	-0.18***	-3.72***	0.69	0.58***
	(-5.14)	(-8.87)	(-0.46)	(0.66)	(11.65)	(-3.43)	(-3.16)	(-4.37)	(0.09)	(12.89)
PPE/A_{t-1}	-0.03	-0.59**	-2.30	-10.47	-0.63	-0.01	0.08	4.71	-109.66	-2.33***
	(-0.74)	(-2.31)	(-0.10)	(-0.24)	(-1.44)	(-0.06)	(0.14)	(0.53)	(-1.31)	(-5.00)
D/A_{t-1}	0.00	1.29***	35.55**	-33.79	0.34	0.18	0.56	7.85	3.47	-0.72*
	(-0.13)	(7.26)	(2.25)	(-1.11)	(1.12)	(1.06)	(1.15)	(1.07)	(0.05)	(-1.87)
Ind. Effects	0.02***	0.04	7.71***	9.29*	0.39***	0.04	0.09	-1.32	17.33	0.39***
	(3.05)	(1.21)	(2.62)	(1.63)	(6.88)	(1.55)	(1.15)	(-1.13)	(1.56)	(6.36)
R^2	0.16	0.20	0.03	0.13	0.31	0.09	0.08	0.05	0.10	0.40
Adj. R^2	0.14	0.19	0.01	0.11	0.30	0.07	0.06	0.04	0.09	0.39
N	447	447	447	447	447	432	432	432	432	432

Panel B: Underpricing

	Stand-alone Firms					Group-affiliated Firms				
	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize	P/A_t	P/A_{t-1}	%Issued	Offer/Book	LnDSize
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Intercept	0.24***	0.92***	25.15***	37.82	14.33***	0.70***	1.74***	46.70***	82.24	15.32***
	(6.06)	(4.76)	(4.11)	(1.11)	(41.61)	(3.02)	(2.58)	(5.30)	(0.86)	(29.09)
Hot	-0.02	-0.08	3.87	-6.27	-0.16	0.15*	0.36	5.86**	51.82	0.52***
	(-1.01)	(-1.04)	(1.52)	(-0.44)	(-1.10)	(1.84)	(1.55)	(1.92)	(1.56)	(2.86)
Mt/B_t	0.01***	0.02	-0.74	27.28***	0.11***	0.00	0.01	-0.04	16.42***	0.03*
	(3.47)	(1.07)	(-1.13)	(7.47)	(3.02)	(0.56)	(0.45)	(-0.12)	(4.89)	(1.87)
$EBITDA/A_{t-1}$	0.18**	1.53***	19.66	191.42***	0.40	-0.98***	-2.99***	-4.11	169.92	-0.87
	(2.19)	(3.85)	(1.56)	(2.73)	(0.56)	(-3.20)	(-3.36)	(-0.35)	(1.34)	(-1.26)
$Lsize_{t-1}$	-0.02***	-0.19***	-2.69***	1.74	0.48***	-0.08***	-0.20***	-3.87***	-3.45	0.55***
	(-5.38)	(-8.98)	(-3.93)	(0.46)	(12.45)	(-3.90)	(-3.55)	(-5.21)	(-0.43)	(12.45)
PPE/A_{t-1}	-0.06	-0.57**	-1.13	-39.65	-0.44	0.00	0.10	2.74	-108.97	-2.28***
	(-1.18)	(-2.45)	(-0.15)	(-0.97)	(-1.07)	(-0.02)	(0.17)	(0.35)	(-1.29)	(-4.91)
D/A_{t-1}	0.00	1.07***	18.86***	-35.27	0.27	0.20	0.61	12.22*	9.99	-0.66*
	(-0.07)	(6.91)	(3.85)	(-1.29)	(0.97)	(1.16)	(1.24)	(1.90)	(0.14)	(-1.71)
Ind. Effects	0.01**	0.04	1.49	7.13	0.33***	0.04	0.08	-1.35	15.50	0.37***
	(2.35)	(1.21)	(1.61)	(1.38)	(6.30)	(1.35)	(0.98)	(-1.31)	(1.38)	(6.07)
R^2	0.13	0.17	0.05	0.14	0.30	0.08	0.08	0.08	0.09	0.40
Adj. R^2	0.12	0.16	0.04	0.13	0.30	0.07	0.06	0.06	0.07	0.39
N	516	516	516	516	516	432	432	432	432	432

Notes: This table reports the market timing effects on the SEO issuance activity of stand-alone and group-affiliated firms using the following regression equation:

$$Y_t = \alpha + \beta_1 HOT + \beta_2 M/B_t + \beta_3 EBITDA/A_{t-1} + \beta_4 Lsize_{t-1} + \beta_5 PPE/A_{t-1} + \beta_6 D/A_{t-1} + D_{Ind} + \varepsilon_t$$

where Y_t equals P/A_t, P/A_{t-1}, %Issued, Offer/Book and LnDSize. t subscript denotes the SEO year. The dummy variable HOT takes the value 1 if the firm issued equity in the hot-issue market and 0 otherwise. Lsize is the log of net sales in millions. D_{Ind} is industry dummy. All the regressions are estimated with industry-fixed effects. All other variables are defined in Table 1. Panel A reports the difference of mean test and regression results when the firms are classified into hot and cold-issue market on the basis of monthly IPO volume and Panel B reports these results when the firms are classified into hot and cold-issue market on the basis of underpricing. *t*-statistics are given in the parentheses and have been estimated using clustered standard errors.

*** statistically significant at 1%

** statistically significant at 5%

* statistically significant at 10%

Table 5: **Calendar Time Regression of IPOs (Monthly Volume)**

Panel A: Stand-alone Firms

	Hot market			**Cold market**			**Difference**		
	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	-0.06***	-0.06***	-0.06***	-0.03	-0.02	-0.02	-0.04**	-0.05**	-0.04**
	(-3.17)	(-2.72)	(-2.70)	(-1.60)	(-1.03)	(-1.07)	(-2.3)	(-2.33)	(-2.16)
R_m-R_f	0.95***	0.95***	0.95***	0.97***	0.98***	0.98***	-0.01	-0.02	-0.01
	(34.54)	(32.84)	(32.56)	(32.65)	(32.06)	(30.9)	(-0.47)	(-0.58)	(-0.31)
SML		-0.02	-0.01		-0.06	-0.07		0.01	0.02
		(-0.43)	(-0.26)		(-1.41)	(-1.46)		(0.42)	(0.69)
HML		0.01	0.01		0	0		-0.01	0
		(-0.14)	(0.24)		(0.01)	(-0.08)		(-0.33)	(-0.09)
Mom			0.03			-0.03			0.1
			(0.34)			(-0.38)			(1.19)
R^2	0.86	0.86	0.86	0.84	0.84	0.84	0.02	0.02	0.01
Adj. R^2	0.86	0.86	0.86	0.84	0.84	0.84	0.01	0.02	-0.01

Panel B: Group-affiliated Firms

	Hot market			**Cold market**			**Difference**		
	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	-0.04***	-0.02	-0.02	-0.08***	-0.07***	-0.07***	0.03	0.04	0.04
	(-2.30)	(-1.23)	(-1.07)	(-4.45)	(-3.82)	(-3.81)	(0.98)	(1.06)	(1.04)
R_m-R_f	0.96***	0.97***	0.98***	0.90***	0.91***	0.91***	0.04	0.05	0.05
	(40.69)	(41.79)	(40.61)	(35.97)	(34.17)	(34.49)	(0.98)	(1.05)	(1.16)
SML		-0.12***	-0.11***		-0.03	0		-0.02	0.01
		(-3.42)	(-2.96)		(-0.76)	(0.05)		(-0.33)	(0.12)
HML		-0.07	-0.05		-0.03	-0.01		0.06	0.08

		(-1.52)	(-1.09)		(-0.81)	(-0.15)		-1.13	(1.42)
Mom			0.09			0.16			0.17
			(0.99)			(2.06)**			(1.06)
R^2	0.92	0.92	0.92	0.87	0.88	0.88	0.01	0.02	0.03
Adj. R^2	0.91	0.92	0.92	0.87	0.87	0.88	0	-0.01	-0.01

Notes: This table reports the calendar-time regression results to examine the long-run performance of stand-alone and group-affiliated IPOs for three years after the IPO listing. It shows the long-run performance of hot and cold IPOs and difference between the long-run performance of hot and cold IPOs. Hot and cold market firms are classified on the basis of monthly IPO volume. Panel A reports the regression results for stand-alone IPOs and panel B reports the regression results for group-affiliated IPOs.

Columns (1) - (6) of the table report the regression results for long-run performance using the following equation:

$$R_{pt} - R_{ft} = \alpha + \beta_1 (R_{mt} - R_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \varepsilon_t$$

Columns (7) - (9) of the table report the regression results of the following regression for the difference in the performance of hot IPOs and cold IPOs:

$$H_{pt} - C_{pt} = \alpha + \beta\,(R_{mt} - R_{ft}) + s\,SMB_t + h\,HML_t + \varepsilon_t$$

In the above equations, R_{pt} is the monthly portfolio returns of the IPOs calculated for the month *t* and R_{ft} is the one year risk-free rate. $(R_{mt} - R_{ft})$ is the market risk premium, where R_{mt} is the market return for the month t, which is COSPI index return in this case. SMB_t is the monthly return on the portfolio of small stocks minus monthly return on the portfolio of large stocks. HML_t is the monthly return on the portfolio of high book-to-market minus the monthly return on the portfolio of low book-to-market returns. The forth factor added by Carhart (1997), MOM_t is the momentum factor which is returns on the portfolio of high momentum stocks (high past returns i.e. winners) minus returns on the portfolio of low momentum stocks (low past returns i.e. losers). Momentum is computed on the basis of previous one year returns. H_{pt} is the monthly portfolio returns of hot stand-alone IPOs for the month *t*. C_{pt} is the monthly portfolio returns of cold stand-alone IPOs for the month *t*. *t*-statistics are given in the parentheses and have been estimated using clustered standard errors.

*** statistically significant at 1%

** statistically significant at 5%

* statistically significant at 10%

Table 6: **Calendar Time Regression of SEOs (Monthly Volume)**

Panel A: Stand-alone Firms									
	Hot market			**Cold market**			**Difference**		
	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart
	1	2	3	4	5	6	7	8	9
Intercept	-0.03	-0.05*	-0.05*	-0.02	0	0	-0.02	-0.02	-0.03
	(-1.25)	(-1.78)	(-1.84)	(-1.40)	(-0.11)	-0.27	(-0.56)	(-0.62)	(-0.76)
R_m-R_f	0.99***	0.98***	0.97***	1.04***	1.06***	1.07***	-0.06	-0.06	-0.07
	(27.01)	(26.15)	(25.02)	(33.48)	(38.88)	(35.57)	(-1.07)	(-1.11)	(-1.22)
SML		0.12**	0.11*		-0.10*	-0.10*		0.04	0.05
		(1.98)	(1.69)		(-1.69)	(-1.73)		(0.36)	(0.38)
HML		0.05	0.03		0.11**	0.11**		-0.09	-0.09
		(0.69)	(0.44)		(2.1)	(2.02)		(-0.72)	(-0.73)
Mom			-0.09			0.09			-0.09
			(-0.61)			(1.03)			(-0.52)
R^2	0.83	0.83	0.83	0.93	0.95	0.95	0.01	0.04	0.04
Adj. R^2	0.83	0.83	0.83	0.92	0.94	0.94	0.01	0.03	0.01
Panel B: Group-affiliated Firms									
	Hot market			**Cold market**			**Difference**		
	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart	CAPM	Fama & French	Carhart
	1	2	3	4	5	6	7	8	9
Intercept	-0.05**	-0.02	-0.01	0.01	0.02	0.03	-0.08***	-0.05**	-0.06*
	(-2.17)	(-0.70)	(-0.42)	(0.38)	(0.5)	(0.8)	(-2.56)	(-2.13)	(-1.79)
R_m-R_f	0.94***	0.98***	1.00***	1.10***	1.10***	1.13***	-0.20***	-0.18***	-0.16*
	(27.46)	(29.72)	(29.99)	(21.83)	(21.27)	(20.71)	(-3.09)	(-2.57)	(-2.21)
SML		-0.18***	-0.15***		-0.01	0.01		-0.19	-0.19
		(-4.06)	(-3.26)		(-0.13)	(0.09)		(-1.38)	(-1.39)
HML		-0.22***	-0.19***		0.08	0.1		-0.25*	-0.25*

		(-4.48)	(-3.79)		-1.09	(1.28)		(-1.78)	(-1.77)
Mom			0.23***			0.17			0.14
			(2.37)			(1.37)			(0.84)
R^2	0.82	0.84	0.85	0.78	0.78	0.78	0.08	0.11	0.12
Adj. R^2	0.81	0.84	0.84	0.78	0.78	0.78	0.07	0.08	0.08

Notes: This table reports the calendar-time regression results to examine the long-run performance of stand-alone and group-affiliated SEOs for three years after the equity issuance. It shows the long-run performance of hot and cold SEOs and difference between the long-run performance of hot and cold IPOs. Hot and cold market firms are classified on the basis of monthly SEO volume. Panel A reports the regression results for stand-alone SEOs and panel B reports the regression results for group-affiliated SEOs.

Columns (1) - (6) of the table report the regression results for long-run performance using the following equation:

$$R_{pt} - R_{ft} = \alpha + \beta_1(R_{mt} - R_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + \varepsilon_t$$

Columns (7) - (9) of the table report the regression results of the following regression for the difference in the performance of hot SEOs and cold SEOs:

$$H_{pt} - C_{pt} = \alpha + \beta\,(R_{mt} - R_{ft}) + s\,SMB_t + h\,HML_t + \varepsilon_t$$

In the above equations, R_{pt} is the monthly portfolio returns of the IPOs calculated for the month *t* and R_{ft} is the one year risk-free rate. $(R_{mt} - R_{ft})$ is the market risk premium, where R_{mt} is the market return for the month t, which is COSPI index return in this case. SMB_t is the monthly return on the portfolio of small stocks minus monthly return on the portfolio of large stocks. HML_t is the monthly return on the portfolio of high book-to-market minus the monthly return on the portfolio of low book-to-market returns. The forth factor added by Carhart (1997), MOM_t is the momentum factor which is returns on the portfolio of high momentum stocks (high past returns i.e. winners) minus returns on the portfolio of low momentum stocks (low past returns i.e. losers). Momentum is computed on the basis of previous one year returns. H_{pt} is the monthly portfolio returns of hot stand-alone IPOs for the month *t*. C_{pt} is the monthly portfolio returns of cold stand-alone IPOs for the month *t*. *t*-statistics are given in the parentheses and have been estimated using clustered standard errors.

*** statistically significant at 1%

** statistically significant at 5%

* statistically significant at 10%